Name:_____

Brain

How do you remember the way to your friend's house? Why do your eyes blink without you ever thinking about it? Where do dreams come from? Your brain is in charge of these things and a lot more.

In fact, your brain is the boss of your body. It runs the show and controls just about everything you do, even when you're asleep. Not bad for something that looks like a big, wrinkly, grey sponge.

Your brain has many different parts that work together. We're going to talk about these five parts, which are key players on the brain team:

cerebrum (say: suh-ree-brum) - *write it out again here* _____

cerebellum (say: sair-uh-bell-um) - *write it out again here* _____

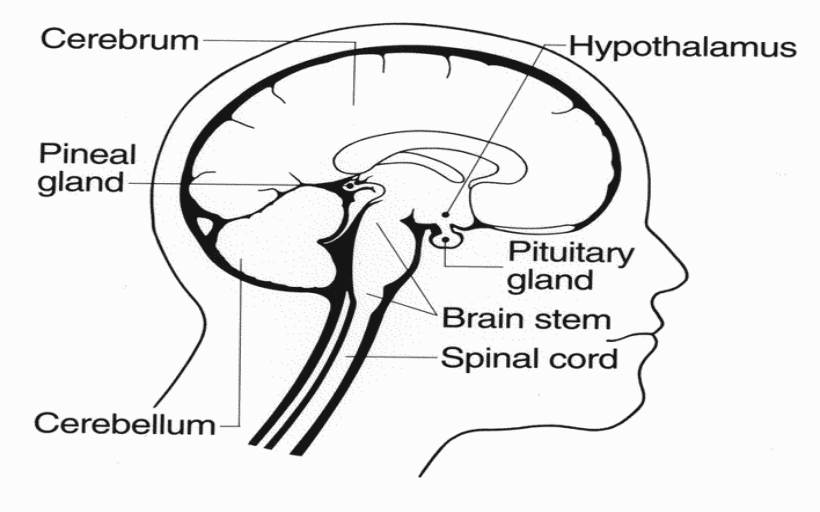
brain stem (or medulla) - *write it out again here* _____

pituitary gland (say: pih-too-uh-ter-ee gland) - *write it out again here* _____

hypothalamus (say: hy-po-thal-uh-mus) - *write it out again here* _____

The Biggest Part: the Cerebrum. (pg 227)

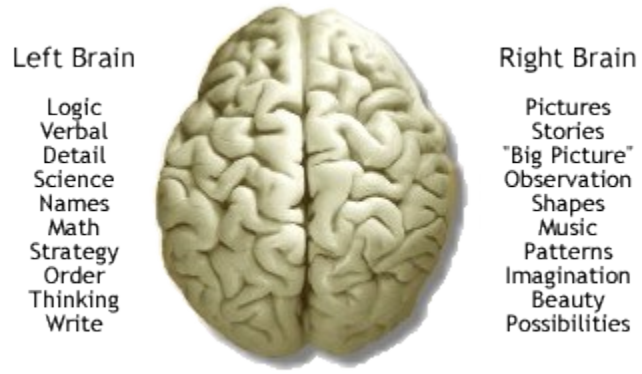
Colour just the cerebrum in on the diagram below.



The biggest part of the brain is the _____. The cerebrum takes up ___ % of the brain's space. The cerebrum is the thinking part of the brain and it controls your _____ thoughts and _____ movements — the ones

that move when you want them to. So you can't dance — or kick a soccer ball — without your cerebrum.

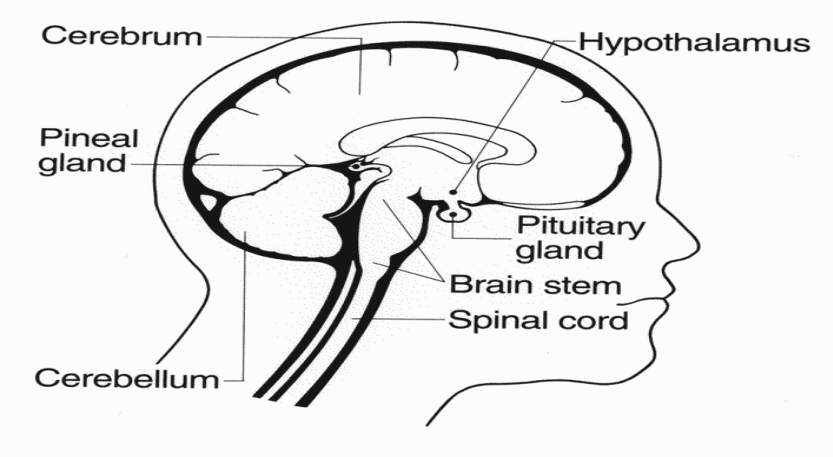
The cerebrum is covered in lots of folds which increase the _____ _____ by three times.



The cerebrum has two halves, with one on either side of the head. Each side is called a _____. Some scientists think that the right half helps you think about abstract things like music, colours, and shapes. The left half is said to be more analytical, helping you with math, logic, and speech. Scientists do know for sure that the right half of the cerebrum controls the _____ side of your body, and the left half controls the _____ side. So if you want to lift your right arm your _____ hemisphere will be doing the work.

The Cerebellum's Balancing Act. (pg 228)

Colour just the Cerebellum on the diagram below.'



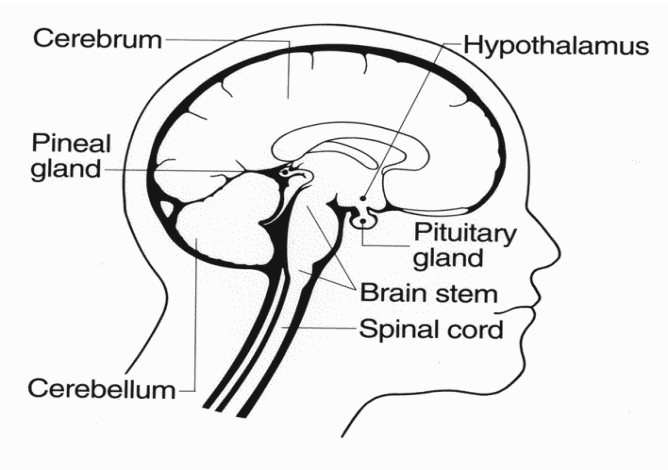
The cerebellum is at the _____ of the brain, below the cerebrum. It's a lot smaller than the cerebrum, but it's a very important part of the brain. It controls _____ and _____ (how your muscles work together).

Because of your cerebellum, you can stand upright, keep your balance, and move around. Think about a surfer riding the waves on his board. What does

he need most to stay balanced? The best surfboard? The coolest wetsuit? Nope — he needs his cerebellum!

Brain Stem **Keeps You Breathing — and More.** (pg 228)

Colour in the brain stem in the diagram below.



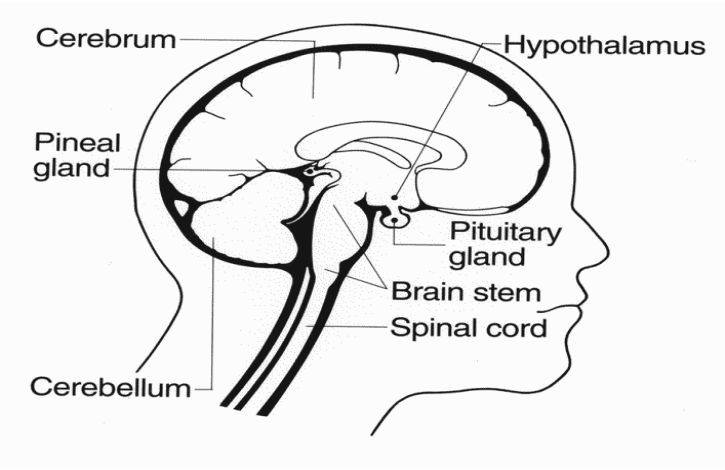
The brain stem is also known as the _____. It sits beneath the cerebrum and in front of the cerebellum. It connects the rest of the brain to the spinal cord, which runs down your neck and back.

The brain stem is in charge of all the _____ functions. Vital means you can not live without it. Things like _____, _____ and _____ rate are vital functions controlled by the _____.

Part of the brain stem's job is to control your involuntary muscles — the ones that work automatically, without you even thinking about it. There are involuntary muscles in the heart and stomach, and it's the brain stem that tells your heart to pump more blood when you're biking or your stomach to start digesting your lunch.

Hypothalamus **Controls Temperature and other things.** (pg 237)

Colour in the hypothalamus in the diagram below.

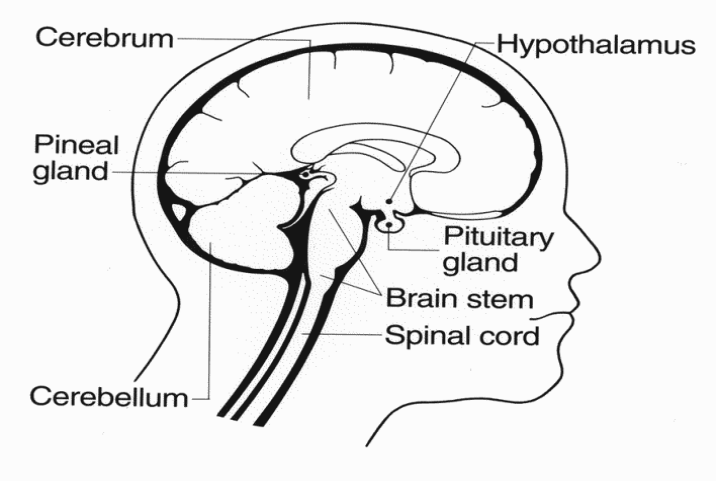


The hypothalamus is part of your brain and is made of _____. It is important because it links the _____ system with the _____ system by receiving nervous impulses and releasing hormones.

The hypothalamus keeps check on your _____ _____. Things like your body temperature and how hydrated you are (how much water is in your body). If any of the conditions in your body are not right your hypothalamus _____ by releasing _____ that affect the _____ gland.

Pituitary Gland **Controls Growth and other things.** (pg 237)

Colour in the pituitary in the diagram below.



The pituitary gland is very small — only about the size of a pea. The pituitary gland detects hormones released by the _____ and releases _____ of its own that affect your body.

Being deep inside your skull means the pituitary gland and hypothalamus are well _____ and less likely to be damaged.

If your clothes from last year are too small, it's because your pituitary gland released hormones that made you grow. This gland is a big player in puberty too. Boys' and girls' bodies go through major changes as they slowly become men and women, all thanks to hormones released by the pituitary gland.

Locate each of the parts of the brain on the diagram below. Use a colour code to identify each and write a few words to describe the function

- ☐ cerebrum
- ☐ cerebellum
- ☐ brain stem
- ☐ pituitary gland
- ☐ hypothalamus

